
Neurophysiological Measures of Involuntary and Voluntary Attention Allocation and Dispositional Differences in Need for Cognition

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Need for cognition (NFC) refers to stable individual differences in the intrinsic motivation to engage in and enjoy effortful cognitive endeavors and has been a useful predictor of dispositional differences in information processing. Although cognitive resource allocation conceptualized as cognitive effort is assumed to be the key mediator of NFC-specific processing, to date no research has systematically addressed its underpinnings. Using a neurocognitive paradigm and recording event-related potentials associated with bottom-up and top-down-driven aspects of attention, the present research contributes to filling this gap. In Study 1, high-NFC individuals showed larger P3a amplitudes to contextually novel events, indicating greater involuntary (automatic) attention allocation. This effect was replicated in Study 2, where NFC also was positively correlated with the P3b to target stimuli, indicating voluntary (controlled) processes of attention allocation. Thus, our findings provide first evidence for neurophysiological correlates of NFC and can improve the understanding of NFC-specific processing.

Keywords: P3a; P3b; event-related potentials; attention allocation; cognitive motivation

Twenty-six years ago, Cacioppo and Petty (1982) introduced the concept of need for cognition (NFC), which refers to dispositional differences in cognitive motivation. Revising and specifying an earlier conceptualization by Cohen, Stotland, and Wolfe (1955), they

proposed NFC to reflect stable individual differences in the intrinsic motivation to engage in and enjoy effortful cognitive endeavors. Indeed, the NFC concept has been an extraordinarily useful descriptor and predictor of individual differences in information processing (for review, see Cacioppo, Petty, Feinstein, & Jarvis, 1996), but there is little evidence of the working mechanisms underlying those differences. This holds particularly for neurocognitive mechanisms of NFC-specific processing, which have not been investigated so far. Therefore, the present report aims at addressing this gap using neurophysiological measures of attention to further specify NFC-specific resource allocation that affects cognitive efficiency. In the following, the conceptualization and known correlates of NFC will be summarized and assumptions on potential neurophysiological correlates of NFC will be derived.

Conceptualization and Behavioral Correlates of NFC

On a bipolar continuum, high NFC refers to the general tendency to engage in and enjoy effortful cognitive activity, whereas low NFC reflects the tendency to avoid

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or minimize effortful cognitive endeavors (Cacioppo et al., 1996). In this regard, a main aspect of individual differences in NFC is the extent to which individuals allocate their cognitive resources to information processing that is conceptualized as *cognitive effort* (Cacioppo & Petty, 1982). In fact, there is considerable empirical evidence that supports cognitive effort as a central source of NFC-specific differences. To examine the link between NFC and effortful processing, a diverse set of measures and procedures has been employed, such as self-reported effort, recall measures, as well as attitudinal and behavioral indicators (for review, see Cacioppo et al., 1996). It has been shown that individuals high in NFC expend more effort in thinking about information, events, arguments, or other people's behavior (e.g., Cacioppo, Petty, & Morris, 1983; Latimer, Katulak, Mowad, & Salowey, 2005; Srull, Lichtenstein, & Rothbart, 1985). At the same time, they are less influenced by peripheral, less-relevant information but elaborate more on issue-relevant information (Haugtvedt, Petty, & Cacioppo, 1992; Towler & Dipboye, 2006). According to the elaboration likelihood model (ELM; Petty & Cacioppo, 1986), these findings can be interpreted in terms of a higher likelihood of high-NFC individuals to engage in deep elaboration, or *central processing*, whereas low-NFC individuals tend to show little elaboration and rather *peripheral processing*. Supporting evidence for effortful and elaborated processing also comes from studies demonstrating that high NFC is associated with better performance in cognitive tasks (Dornic, Ekehammar, & Laaksonen, 1991; Sadowski & Gulgoz, 1992), with greater success in adaptive decision making (Levin, Huneke, & Jasper, 2000; Sargent, 2004) and problem solving (e.g., Nair & Ramnarayan, 2000) as well as with better performance in complex skill acquisition (Day, Espejo, Kowollik, Boatman, & McEntire, 2007).

In addition to showing more effective behavior in a variety of tasks, high-NFC individuals enjoy cognitive activities even in the absence of external reward or performance feedback (Thompson, Chaiken, & Hazlewood, 1993), prefer complex to simple tasks (Cacioppo & Petty, 1982), and exhibit a stronger propensity to search for new information (Das, Echambadi, McCardle, & Luckett, 2003; Verplanken, Hazenberg, & Palenewen, 1992).

Consistent with this, efforts to relate NFC to established personality dimensions such as the five factors proposed by Costa and McCrae (1992) have shown that NFC is positively correlated with Conscientiousness and Openness to Experience (Sadowski & Cogburn, 1997; Tuten & Bosniak, 2001; Woo, Harms, & Kuncel, 2007), whereas concerning an association between NFC and cognitive ability, no or only moderate correlations have been observed (Bors, Vigneau, & Lalande, 2006; Cacioppo,

Petty, Kao, & Rodriguez, 1986; Day et al., 2007; Tidwell, Sadowski, & Pate, 2000). Integrating these findings with the results outlined above, individuals high in NFC tend to show stronger intrinsically motivated goal-oriented behavior and, at the same time, are more attracted by complexity and novelty, without necessarily having a higher ability to achieve their goals.

To summarize, cognitive effort has been considered as the core aspect of individual differences in NFC and has been conceptualized as a dimension influencing the elaboration likelihood of information and the depth of processing, respectively (Cacioppo et al., 1986; Craik & Lockhart, 1972; Petty & Wegener, 1999). So far, NFC-specific cognitive effort has been investigated mainly by means of a posteriori measures such as self-reported effort, information recall, and attitudinal indicators. However, because attention allocation is regarded as a crucial precondition of higher level cognition (e.g., Cowan, 1988; Fisk & Schneider, 1984; Parasuraman, 2000), including the depth of processing (e.g., Craik, 2002) and thus the elaboration likelihood of information, we suggest attentional resource allocation to be an important factor of cognitive effort or effortful processing regarding NFC. This notion is further supported because environmental stimuli that do not receive full attention and are analyzed only to a shallow sensory level give rise to very transient encoding, whereas stimuli that are attended to and fully analyzed result in generally deeper encoding (Craik & Tulving, 1975).

In the following, a neurocognitive paradigm is introduced that allows us to examine temporal and functional aspects of attentional resource allocation via event-related brain potentials (ERPs). Because this approach yields insights into involuntary and voluntary (or automatic and controlled) aspects of resource allocation, it provides an appropriate, and more direct online measure to further specify cognitive effort or effortful processing in terms of NFC.

Auditory Novelty Oddball and Event-Related Brain Potentials

The auditory novelty oddball paradigm is established to study attention-related processes using neurophysiological methods, for example, the electroencephalogram (EEG, e.g., Debener, Kranczioch, Hermann, & Engel, 2002; Fabiani & Friedman, 1995). In the paradigm, three classes of auditory events are presented: a frequent *standard* event; a rare *target* event; and an equally rare set of unique events, so-called *novels*, mostly consisting of environmental sounds (e.g., animal, human, and machine noises). Whereas the participant's voluntary attention is directed to the target tones that demand a particular response (e.g., counting the targets), the participant's

involuntary attention is assumed to be captured by the task-irrelevant, but highly salient, novel sounds.

Because of their excellent temporal resolution, ERPs provide a sensitive measure to assess the time course of information processing and thus permit us to study neuronal activity patterns associated with perceptual and cognitive processes in real time (Debener, Makeig, Delorme, & Engel, 2005; Friedman, Cycowicz, & Gaeta, 2001). One of the most prominent ERP components, the P300, has demonstrated considerable utility in the study of many aspects of cognition, such as attention, perception, and memory (e.g., Fabiani, Karis, & Donchin, 1986; Polich & Kok, 1995), as well as for assessing a variety of mental disorders, such as dementia (e.g., Frodl et al., 2002) or schizophrenia (for review, see Ford, 1999). Two functionally distinct P300 components can be discriminated, the P3a and the P3b (Friedman et al., 2001; Polich, 2003, 2007), which are introduced in the following paragraphs.

The P3a component. In the auditory domain, contextually deviant or novel stimuli evoke a positive deflection in the ERP peaking between 250 ms and 350 ms after stimulus onset with a frontocentral maximum. This positivity is commonly labeled as *P3a* or—if evoked by novel sounds—as *novelty P3* (Courchesne, Hillyard, & Galambos, 1975; Simons, Graham, Miles, & Chen, 2001). It has been repeatedly shown that the P3a amplitude shows habituation (e.g., Friedman & Simpson, 1994). For this reason, the P3a often is discussed as indicating a neurophysiological correlate of the orienting response (Sokolov, 1963), that is, an involuntary attentional switch to contextually deviant events of potential behavioral significance (e.g., Cycowicz & Friedman, 1998; Näätänen, 1990). Therefore, the P3a has been suggested to mirror involuntary (automatic), bottom-up-driven attentional processes (Daffner et al., 2000; Knight, 1984, 1991). Furthermore, the component is assumed to reflect an evaluative, precategorizational aspect of orienting as opposed to the detection of deviance per se. In this context, the P3a is seen as indicating aspects of resource allocation within frontal attentional networks (Friedman et al., 2001; Holdstock & Rugg, 1995), with the P3a amplitude being positively correlated with the degree of attentional resources allocated to the processing of salient stimuli (e.g., Daffner et al., 2003; Debener et al., 2005).

Escera, Alho, Winkler, and Näätänen (1998) delineated two P3a subcomponents of presumably different functionality—a centrally distributed *early P3a* peaking around 230 ms and a *late P3a* of right frontal scalp distribution peaking around 315 ms after stimulus onset (see also

Figure 1). They proposed that the late P3a more closely reflects the actual reorientation of attention, whereas the early P3a might reflect a violation of a polysensorial model of the external world maintained in the temporal-parietal association cortex (Escera et al., 1998; Yago, Escera, Alho, Giard, & Serra-Grabulosa, 2003).

The P3b component. The P3a is followed by the spatiotemporally partly overlapping, but parietally distributed, *P3b*, which peaks after 400 ms. This ERP component is typically pronounced for target stimuli about which participants are required to generate a certain response. For that reason, the target-evoked P3b also is referred to as *target P3* (e.g., Debener et al., 2002; Donchin, Spencer, & Dien, 1997).

Contrary to involuntary attentional processes indicated by the P3a component, the P3b deflection is related to voluntary (controlled), top-down-driven aspects of attentional resource allocation (e.g., Debener et al., 2002; Kok, 2001). Here, the P3b is considered a neuronal correlate of stimulus evaluation and categorization, indicating the extent to which a stimulus matches an existing or precategorized working memory template (for review, see Kok, 2001). This approach is widely paralleled by the so-called context updating hypothesis, suggesting that the P3b mirrors a need to revise a current representation maintained in working memory, with the P3b being positively correlated with the extent of working memory updating (e.g., Donchin & Coles, 1988; Polich, 2003). The updating of representations in working memory is governed by attentional processes (Polich, 2004, 2007) and is supposed to be an important mechanism underlying the consolidation of new information and the facilitation of subsequent recall (Fabiani & Donchin, 1995; Johnson, 1995). In sum, the P3b (or target P3) is interpreted as reflecting the amount of attentional resources voluntarily invested in stimulus processing (Debener et al., 2002; Kok, 2001; Polich, 2003).

Event-Related Brain Potentials, Resource Allocation, and NFC

Given that the P3a and P3b provide insights into functional and temporal aspects of bottom-up and top-down-driven aspects of attentional resource allocation, the present approach offers a more direct and objective measure to study cognitive mechanisms underlying NFC-specific processing.

Because high-NFC individuals engage in and enjoy cognitive activities, are more conscientious, and are therefore more inclined to willingly invest cognitive resources in issue-relevant information processing (see Cacioppo et al., 1996), we particularly hypothesize NFC-specific differences in voluntary, top-down-driven

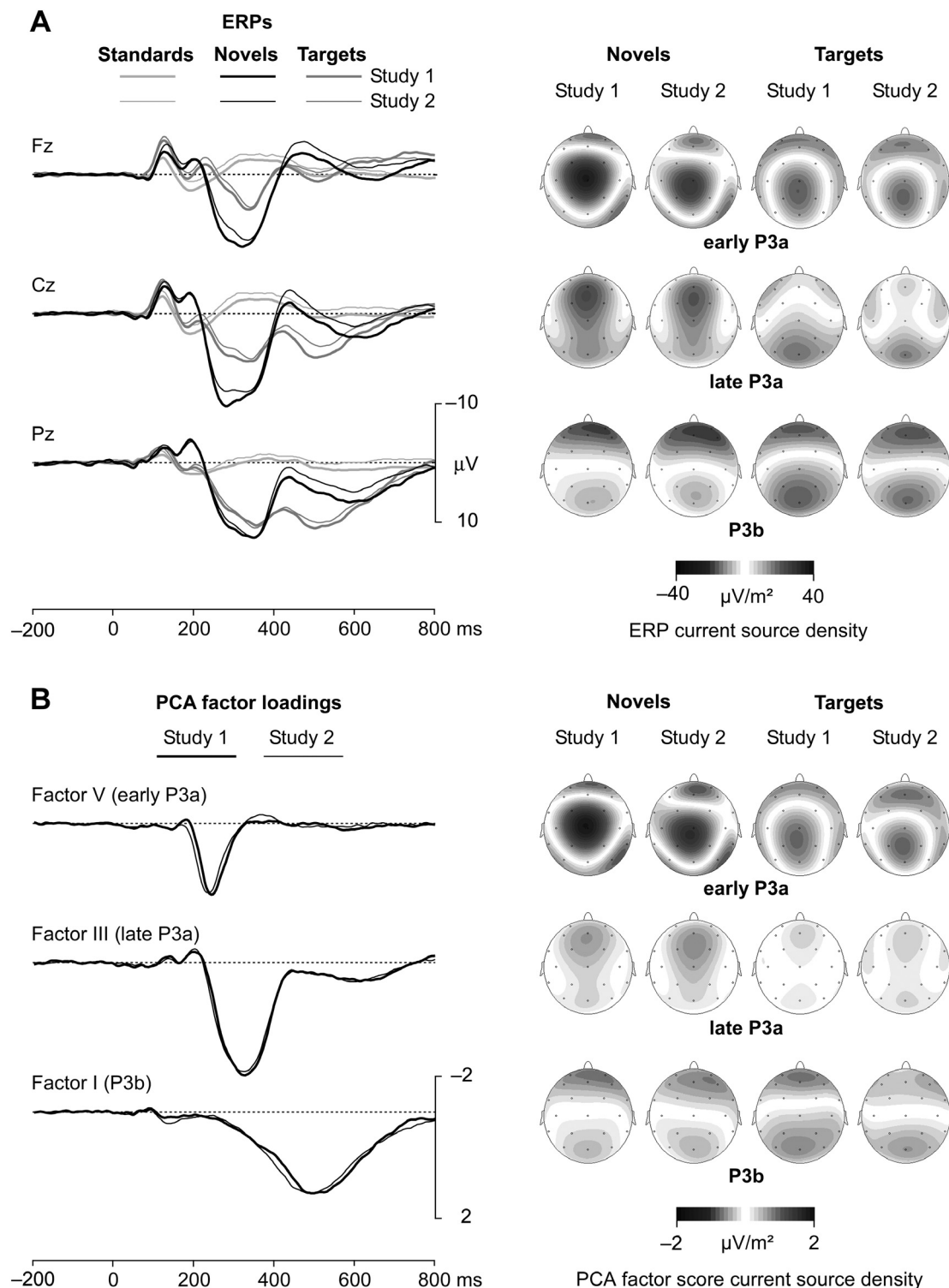


Figure 1 (A) Left panel: midline grand mean event-related brain potentials (ERPs) to standards (light gray lines), novels (black lines), and targets (gray lines) for Study 1 (thick line) and Study 2 (thin line); right panel: current source density maps of the ERPs temporally corresponding to the principal components analysis (PCA) factors, separately for the novel and target condition in Studies 1 and 2 (maps for standards are not displayed due to the absence of P3 components in this condition, so the maps would essentially remain blank); (B) left panel: rotated factor loadings of the PCA factors corresponding to early P3a (peaking at 240 ms in Study 1 and at 232 ms in Study 2), late P3a (peaking at 324 ms in Study 1 and 320 ms in Study 2), and P3b (peaking at 492 ms in both studies); right panel: current source density maps of the respective factor scores separately for the novel and target condition in Studies 1 and 2.

attentional processing of target stimuli as indicated by the P3b. However, given the tendency of high-NFC individuals to search for new information and their higher scores in Openness to Experience, it is also reasonable to assume that NFC may be positively correlated with the P3a, or novelty P3, which mirrors attentional resource allocation to contextually novel events of potential behavioral significance. This directing of attention to sudden changes in the environment and the resulting involuntary stimulus evaluation and precategorization facilitate conscious interpretation and appropriate action (Daffner et al., 1998; Näätänen, 1992). The notion that NFC might be positively associated with such bottom-up-driven aspects of stimulus processing also is paralleled by the assertion that individuals high in NFC are characterized “by active, exploring minds and, through their senses and intellect, reach and draw out information from their environment” (Cacioppo et al., 1996, p. 199). Taken together, the neurophysiological markers outlined above provide an intriguing opportunity for a better understanding of functional and temporal aspects of NFC-specific resource allocation. The current research addressed these issues in two independent studies.

STUDY 1

To probe possible differences in voluntary and involuntary attentional resource allocation, an auditory novelty oddball paradigm was employed. We expected NFC to be related primarily to voluntary target-processing as indicated by P3b amplitude. Furthermore, we explored a possible association of NFC with involuntary attentional processes as indicated by the P3a to novel sounds.

Method

Participants. The study was based on data from a sample of students of the Dresden University of Technology, which had been examined previously within another context (Strobel et al., 2004). The sample comprised 72 right-handed students (26 men, age $M \pm SD$ 22.3 $\pm$ 2.8 years, range 18–30 years). Participants gave written informed consent and received either monetary compensation or course credit for their participation. Two participants were excluded from the sample due to missing values in NFC.

All participants confirmed to be free of relevant health problems, that they never had received psychopharmacological treatment, and did not abuse drugs. Normal hearing was ensured using an audiometer. Possible confounding factors, such as participants' sleep duration as well as their nicotine, caffeine, and alcohol consumption during the past 24 hours, were assessed via self-report.

Procedure. Participants were seated in a dimly lit, acoustically shielded room and the electrode cap was attached. The recording session started with an 8-min resting EEG, which was part of another study and had no relevance for our research but provided the opportunity to familiarize subjects with EEG procedure. The resting EEG was followed by the auditory novelty oddball paradigm. Within this paradigm, two sine tones of 350 Hz and 650 Hz served as standard (80% probability) and target stimuli (10% probability), respectively. The novel stimuli (10% probability) were taken from a set of environmental sounds established by Fabiani and Friedman (1995) and each environmental sound was presented only once in pseudo-randomized order. The stimuli ($N = 500$) had a mean duration of 320 ms and were presented with a variable interstimulus interval ranging from 950 ms to 1,250 ms. A short break was implemented after 250 stimuli. Participants were requested to silently count the target tones.

EEG recordings and ERP analysis. EEG, vertical electrooculogram (VEOG), and horizontal electrooculogram (HEOG) were continuously recorded (band-pass 0.1–100 Hz, sampling rate 500 Hz) from 32 sites of the enhanced 10–20 system (Ag/AgCl electrodes), affixed to an electrode cap, using the nose-tip as reference and AFz as ground. Electrode impedances were kept below 10 k Ω at all sites. Continuous EEGs were epoched from –200 ms to 800 ms after stimulus onset. Epochs were submitted to an infomax independent component analysis (ICA) to remove artifactual information using EEGLAB (Delorme & Makeig, 2004). In addition, ICA-corrected epochs with values that exceeded the threshold of ± 100 mV were rejected, which resulted in a median percentage of rejected trials of about 4%. The first and the second half of the epochs of each condition were averaged separately (blocks 1 and 2). To linearly decompose the ERP voltage signal, the averages were submitted to an unrestricted, unscaled temporal principal components analysis (PCA) based on the covariance matrix, followed by varimax rotation. Decomposed ERP measures commonly provide a higher reliability and a lower degree of overlap than peak-based amplitudes and, therefore, proved to be better suited for the analysis of spatially and temporally partly overlapping ERP components (e.g., Kayser & Tenke, 2003).¹

The first 10 factors (Eigenvalues: 1641.8, 652.5, 331.9, 301.9, 165.8, 104.0, 91.3, 72.4, 57.7, and 53.1) were interpreted based on their spatiotemporal correspondence with the grand mean ERP. Three factors of the P3 family were identified. Factor I, accounting for 24.3% of the variance and peaking at 492 ms after stimulus

onset with a posterior distribution, was identified as P3b. Factor III (26.8%) showed a frontocentral distribution with a peak at 324 ms, whereas factor V (5.1%) was centrally distributed with a peak at 240 ms. In accordance with previous findings (Escera et al., 1998), these two factors were referred to as late P3a and early P3a. Figure 1A shows the midline grand mean ERPs to standards, novels, and targets as well as the current source density maps of the ERPs temporally corresponding to the PCA factors for novels and targets. The m-shaped ERP pattern in the target condition (Figure 1A, left) shows a small P3a in response to target stimuli because processes of automatic reorientation also occur in conjunction with target events that are sufficiently deviant (Friedman et al., 2001). Figure 1A (left) further shows electrocortical activity of auditory-evoked potentials associated with earlier and later stages of cognitive processing. Figure 1B contains the rotated factor loadings of the PCA factors corresponding to early P3a, late P3a, and P3b as well as the current source density maps of the respective factor scores for novels and targets. To demonstrate equivalence of the PCA components in Study 1 with those of the replication study described below, Figure 1 gives the respective data also for Study 2.

NFC. NFC was assessed using the short version of the German NFC scale (Bless, Wänke, Bohner, Fellhauer, & Schwarz, 1994). The NFC score varied between -20 and +37 ($M = 17.90$, $SD = 11.51$, Cronbach's $\alpha = .85$).

Statistical analysis. First, the component characteristics of early P3a, late P3a, and P3b evoked by novel and target stimuli were separately analyzed in a 2 (stimulus type: targets, novels) $\times$ 2 (block: first and second half of the epochs of each condition) $\times$ 3 (scalp site: Fz, Cz, Pz) repeated-measures ANOVA for each component.

To elucidate the relationship between NFC and the PCA-derived components, the continuous NFC variable was entered into the ANOVA model. Because only small influences of personality dimensions on P3 measures will be discerned in samples of homogeneous university students (Polich & Martin, 1992), we further used an extreme group approach because it has the distinct advantage of providing substantially greater statistical power (Cohen, 1988). According to this, participants were categorized into a high-NFC group, defined as the upper quartile of the NFC scores ($NFC > 26$, $n = 15$), and a low-NFC group, defined as the lower quartile of the NFC scores ($NFC < 12$, $n = 17$).

Due to the fact that the P3a is pronounced for novels and the P3b for target stimuli, factor scores of the P3a subcomponents were analyzed for novel stimuli only, whereas factor scores of the P3b were analyzed for target

stimuli only, resulting in a 2 (block) $\times$ 3 (scalp site) repeated-measures ANOVA for each component. However, because NFC-specific differences in P3a to target and P3b to novel stimuli cannot be ruled out, separate analyses were conducted and possible effects are reported. Degrees of freedom were Greenhouse-Geisser corrected where appropriate. Sex and age were considered as covariates if significantly correlated with NFC or with P3 factor scores. No significant effects of sleep duration, nicotine, caffeine, and alcohol consumption on NFC or P3 factor scores were observed (all $ps > .05$).

Results

ERP characteristics. The early P3a was pronounced in response to novel stimuli (stimulus type main effect), $F(1, 71) = 50.13$, $p < .001$, $\eta^2 = .41$, and showed a central distribution (scalp site main effect), $F(1.5, 105.2) = 45.85$, $p < .001$, $\eta^2 = .39$; planned comparisons: $Cz > Fz/Pz$, $Pz > Fz$, $p < .05$. Similar results were obtained for the frontocentrally maximal late P3a (stimulus type), $F(1, 71) = 104.18$, $p < .001$, $\eta^2 = .60$; (scalp site), $F(1.4, 102.3) = 9.01$, $p < .001$, $\eta^2 = .113$; $Fz/Cz > Pz$, $p < .05$, which also showed habituation with repetition (block main effect), $F(1, 71) = 9.77$, $p = .003$, $\eta^2 = .12$. The P3b component was primarily evoked by target stimuli (stimulus type main effect), $F(1, 71) = 131.04$, $p < .001$, $\eta^2 = .65$, and showed a parietal distribution (scalp site main effect), $F(1.4, 98.8) = 373.66$, $p < .001$, $\eta^2 = .84$; $Pz > Cz > Fz$, $p < .05$. Hence, the component characteristics of the P3a and P3b ERPs clearly correspond to previous findings (e.g., Comerchero & Polich, 1999; Escera et al., 1998).

P3a and NFC. In response to novel stimuli, neither an NFC main effect (continuous variable, $p = .750$, $\eta^2 < .01$; extreme group variable, $p = .550$, $\eta^2 = .01$) nor an NFC-specific interaction effect (all $p > .05$, $\eta^2 \leq .09$) was observed for the early P3a component (Figure 2, left panels). Likewise, no main or interaction effect of the continuous NFC variable emerged for the late P3a component (main effect, $p = .115$, $\eta^2 = .04$; interaction effects, all $ps > .05$, $\eta^2 \leq .01$; Figure 2A, middle panel). However, a significant main effect of NFC appeared if the extreme group variable was entered into the model, $F(1, 29) = 4.35$, $p = .046$, $\eta^2 = .13$. As depicted in Figure 2B (middle panel), high-NFC participants showed significantly larger late P3a amplitudes ($M = 0.27$, standard error of the mean [SEM] = 0.04) than did low-NFC participants ($M = 0.15$, SEM = 0.04). To control the possible influence of a residual overlap of the late P3a with the early P3a and the P3b even after PCA-decorrelation (Debener et al., 2005), we separately averaged the factor scores of early P3a and P3b in response to novel stimuli across midline positions and blocks and included these two component

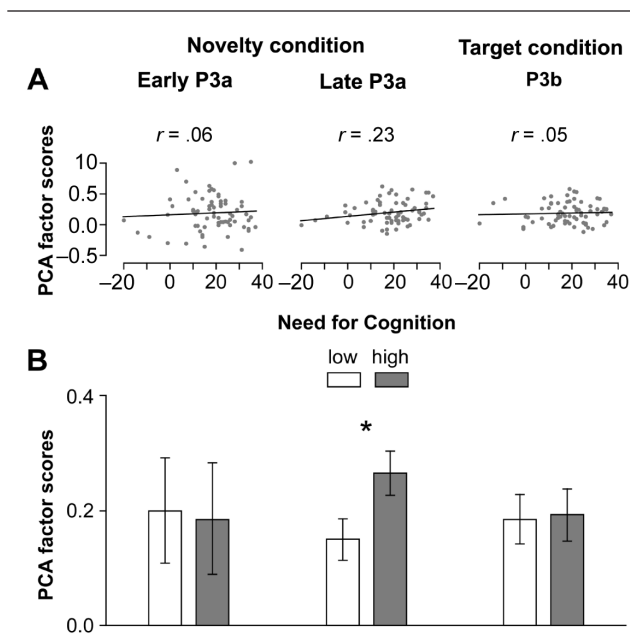


Figure 2 Study 1: Principal components analysis (PCA) factor scores of P3 components, averaged across the midline positions (Fz, Cz, Pz). (A) PCA factor scores plotted against the continuous need for cognition (NFC) variable for early P3a (left) and late P3a (middle) evoked by novel stimuli and P3b (right) evoked by target stimuli; (B) PCA factor scores grouped by low ($n = 15$) versus high NFC ($n = 17$); error bars indicate standard error of the mean (SEM).

* $p < .05$.

averages as additional covariates in the ANOVA model. The NFC effect remained unchanged, $F(1, 27) = 4.40$, $p = .046$, $\eta^2 = .14$.

P3b and NFC. Neither a significant NFC main effect (continuous variable, $p = .722$, $\eta^2 < .01$; extreme group variable, $p = .973$, $\eta^2 < .01$; Figure 2, right panels) nor NFC-specific interaction effects emerged for the target-related P3b component (all $ps > .05$, $\eta^2 \leq .03$).

Behavioral data. On average, participants reported 96.7% ($SD = 5.37$) of the correct target number. NFC had no effect on target count ($p > .05$).

Discussion

Our findings reveal NFC-specific differences in attentional resource allocation, particularly for involuntary, bottom-up-driven stimulus processing. Individuals high in NFC showed significantly larger factor scores of the late P3a component in response to novel stimuli than did low-NFC individuals. By contrast, no NFC-specific differences occurred for the early P3a component. However, the early P3a has been suggested to be a neuronal correlate of a detection system indicating

deviation rather than attentional reorientation, as indicated by the late P3a (Escera et al., 1998; Yago et al., 2003). As outlined above, the late P3a is assumed to mirror the allocation of attentional resources to potentially significant events, thereby facilitating their evaluation and precategorization to form a stimulus-specific representation (e.g., Friedman et al., 2001). Hence, the present findings suggest a more effective attentional monitoring system for high-NFC individuals with positive influence on, for instance, stimulus encoding and appropriate action. If replicated, the present finding would have crucial implications for the specification of cognitive effort in terms of NFC because it suggests that NFC-specific resource allocation is already occurring at an involuntary level of information processing.

Contrary to our assumptions, no association of NFC with top-down-driven aspects of target processing indicated by P3b (or target P3) variation was observed. Given that the P3b mirrors voluntary resource allocation (e.g., Kok, 2001), a central assumption derived from NFC theory and corresponding findings was that high-NFC individuals should focus on target stimuli and therefore allocate a greater amount of attentional resources into target processing, which should be reflected in a larger P3b. However, the null result does not inevitably reflect a lack of NFC-specific differences in top-down-driven resource allocation because a potential limitation of the present paradigm concerns its task difficulty. Counting and processing rare target tones that are easily discriminable from the standard tones and the novel sounds may not have been demanding enough to reveal substantial differences between high- and low-NFC individuals. In this context, it also is conceivable that high-NFC individuals have been more receptive to the novel environmental sounds because these stimuli might have represented a cognitive challenge for them in the otherwise less-demanding task context.

STUDY 2

With Study 2, we attempt to replicate the result of NFC-specific differences in bottom-up-related resource allocation to novel events, as revealed in Study 1. Therefore, the participants carried out the same novelty oddball paradigm. A larger sample was used to increase statistical power and the precision of the estimates. To further investigate top-down-driven, voluntary aspects of attentional resource allocation and NFC, the total number of stimuli was doubled to enhance cognitive task demand without jeopardizing the comparability between both studies. This slight modification was supposed to require participants to maintain attention to the presented stimuli throughout a prolonged period

of time and, through higher task demand, to potentially reveal substantial differences in target processing between high- and low-NFC individuals.

Method

Participants. Ninety-four, right-handed, undergraduate, psychology students of the Dresden University of Technology (20 men, age $M \pm SD$ 22.0 ± 4.3 years, range 18–42 years) participated for course credit. Ten participants were excluded from the sample because of missing NFC or EEG data. As in Study 1, all participants confirmed to be free of relevant health problems, that they never had received psychopharmacological treatment, and did not abuse drugs. Normal hearing was ensured using an audiometer. Participants' sleep duration; their nicotine, caffeine, and alcohol consumption during the past 24 hours; as well as oral contraceptive intake were assessed to control for potential influences on the ERPs. All participants gave written informed consent.

Procedure. The novelty oddball paradigm used in Study 2 was basically consistent with those of Study 1. Only the number of presented stimuli was doubled ($N = 1,000$) without changing stimulus probability (standard stimuli: 80%, target stimuli: 10%, novel stimuli: 10%). Furthermore, being part of a larger research project, Study 2 had a somewhat different arrangement of the experimental setting. Besides the 8-min resting EEG, which always was recorded first, two additional behavioral tasks were conducted prior to and after the novelty oddball paradigm in a randomized order. Because these tasks are not relevant for the present study they are not further explained. It is, however, important to note that the two tasks were of different complexity and therefore may have had a different impact on the novelty oddball performance (see Statistical Analyses). Subsequent to the novelty oddball paradigm, participants completed the German versions of Costa and McCrae's (1992) revised NEO Personality Inventory (NEO-PI-R, German; Ostendorf & Angleitner, 2003), Zuckerman's (1979) Sensation Seeking Scale (SSS-V, German; Beauducel, Brocke, Strobel, & Strobel, 1999), as well as Cloninger's (1987) Tridimensional Personality Questionnaire (TPQ, German; Weyers, Krebs, & Janke, 1995).

EEG recordings and ERP analysis. EEG recordings and ERP analyses corresponded to those of Study 1 and a comparison of the signal-to-noise (SNR) ratio in the two studies, estimated using an algorithm by Möcks, Gasser, and Köhler (1988), showed that despite the higher number of trials in Study 2, the SNR was comparable (ANOVA, $p > .05$) and, descriptively, was even lower in Study 2. This was probably due to some technical problems with

the recording device in Study 2, a notion that is also underscored by a higher median percentage of invalid trials after artifact correction and rejection (about 13% compared to 4% in Study 1). Nevertheless, with a median number of trials available for averaging of 87 (minimum 44, maximum 99) per condition, we had still enough trials for proper ERP analyses. In Study 2, an almost identical factor solution emerged (see Figure 1). Among the first 10 PCA factors (Eigenvalues: 1768.9, 714.6, 266.7, 252.9, 156.6, 92.1, 74.6, 57.2, 40.3, and 36.1), Factor I, accounting for 26.0% of the variance, was identified as P3b, peaking at 492 ms. Factor III (27.0%), peaking at 320 ms, was interpreted as late P3a, whereas Factor V (5.2%), showing a maximum at 232 ms, was referred to as early P3a.

NFC. NFC scores varied between -19 and $+40$ ($M = 13.63$, $SD = 10.73$, Cronbach's $\alpha = .83$).

Statistical analysis. A test of the ERP characteristics was conducted and the relationship between NFC and the ERPs was examined using repeated-measures ANOVAs with the continuous as well as with the extreme group scores of NFC as an independent variable. As in Study 1, participants within the upper (NFC > 20 , $n = 22$) and lower quartile (NFC < 8 , $n = 21$) of the NFC scores were included for analyses with the extreme group variable. The ANOVA models largely corresponded to those outlined in Study 1. In addition, the order of the two behavioral tasks prior to and after the novelty oddball paradigm (see Procedure) was controlled by considering task order as additional between-subjects factor. To ensure that possible ERP associations are specific to NFC, separate ANOVA models were conducted concerning the personality traits Novelty Seeking and Sensation Seeking as independent variables because novelty/sensation seeking behavior is assumed to be positively linked to the P3a ERP (e.g., Daffner, Mesulam, Cohen, & Scinto, 1999; Ratsma, van der Stelt, Schoffelmier, Westerveld, & Gunning, 2001). The NEO factor Extraversion was further considered due to reported associations with the P3b amplitude (e.g., Cahill & Polich, 1992). In addition, possible effects of the NEO factors Conscientiousness and Openness to Experience on the P3 components were investigated because small to moderate positive correlations between NFC and these traits have been consistently reported (Sadowski & Cogburn, 1997; Tuten & Bosniak, 2001; Woo et al., 2007).

Results

ERP characteristics. As expected and similar to Study 1, the early P3a was pronounced in response to novel stimuli (stimulus type main effect), $F(1, 93) = 37.72$, $p < .001$, $\eta^2 = .29$, with a central distribution (scalp site main effect), $F(1.7, 157.8) = 60.88$, $p < .001$, $\eta^2 = .40$;

planned comparisons: $Cz > Fz/Pz$, $Pz > Fz$, $p < .05$. Similar results were obtained for the frontocentrally maximal late P3a (stimulus type), $F(1, 93) = 129.01$, $p < .001$, $\eta^2 = .58$; (scalp site), $F(1.4, 134.1) = 7.39$, $p = .003$, $\eta^2 = .07$; $Cz > Fz/Pz$, $p < .05$, which also showed habituation with repetition (block main effect), $F(1, 93) = 81.14$, $p < .001$, $\eta^2 = .47$. The P3b component was primarily elicited by target stimuli (stimulus type main effect), $F(1, 93) = 167.87$, $p < .001$, $\eta^2 = .64$, and showed a parietal distribution (scalp site main effect), $F(1.3, 124.1) = 284.19$, $p < .001$, $\eta^2 = .75$; $Pz > Cz > Fz$, $p < .05$.

P3a and NFC. Using the continuous NFC variable, no NFC-specific main or interaction effect was observed for the early P3a component in response to novel stimuli (Figure 3A, main effect, $p = .129$, $\eta^2 = .03$; interaction effects, all $ps > .05$, $\eta^2 \leq .01$). For the extreme scores, a marginally significant main effect of NFC emerged, $F(1, 369) = 3.29$, $p = .078$, $\eta^2 = .08$. As illustrated in Figure 3B (left panel), high-NFC participants ($M = 0.14$, $SEM = 0.07$) showed larger early P3a amplitudes than did low-NFC participants ($M = -0.05$, $SEM = 0.07$). However, when statistically controlling for the influence of the adjacent components by entering the averaged scores of the late P3a and P3b as covariates into the ANOVA model, the observed effect of NFC on the early P3a component failed to reach significance, $F(1, 34) = 2.11$, $p = .156$, $\eta^2 = .06$.

As in Study 1, a significant main effect of NFC was observed with respect to the late P3a component, $F(1, 38) = 5.81$, $p = .021$, $\eta^2 = .13$, with larger late P3a factor scores for high-NFC ($M = 0.27$, $SEM = 0.04$) than for low-NFC ($M = 0.15$, $SEM = 0.04$) participants (see Figure 3B, middle panel). In contrast to Study 1, this main effect was not limited to the extreme group design but also was observed for the continuous NFC variable, $F(1, 84) = 7.69$, $p = .007$, $\eta^2 = .09$ (Figure 3A, middle panel). When statistically controlling for the potential influence of the adjacent components, this effect remained unaffected (continuous variable), $F(1, 82) = 7.05$, $p = .010$, $\eta^2 = .08$; (extreme group variable), $F(1, 36) = 5.44$, $p = .025$, $\eta^2 = .13$. Moreover, ANOVA with Conscientiousness as an independent variable revealed a significant effect on the late P3a component, $F(1, 81) = 6.01$, $p = .016$, $\eta^2 = .07$. However, this effect disappeared when the continuous NFC variable was added to the model, $F(1, 76) = 1.46$, $p = .230$, $\eta^2 = .02$, whereas NFC still affected the late P3a amplitude, $F(1, 76) = 4.50$, $p = .037$, $\eta^2 = .06$. None of the other personality dimensions was significantly associated with the late P3a amplitude (all $ps > .05$).

P3b and NFC. In response to target events, no main or interaction effect of the continuous NFC variable

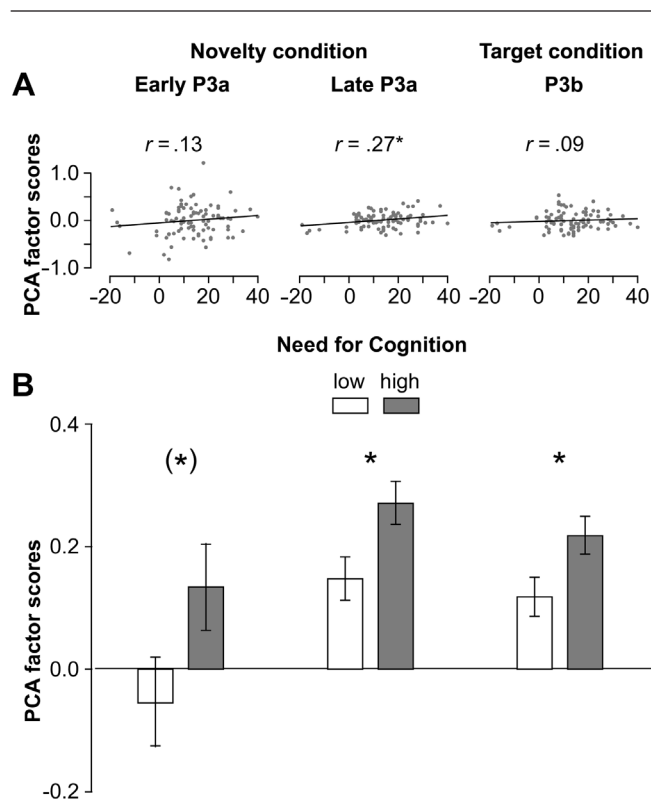


Figure 3 Principal components analysis (PCA) factor scores of P3 components, averaged across the midline positions (Fz, Cz, Pz). (A) PCA factor scores plotted against the continuous need for cognition (NFC) variable for early P3a (left) and late P3a (middle) evoked by novel stimuli and P3b (right) evoked by target stimuli; (B) PCA factor scores, residualized for the order of the behavioral tasks prior and after the novelty oddball, grouped by low ($n = 21$) versus high NFC ($n = 22$); error bars indicate standard error of the mean (SEM).

(*) $p < .10$. * $p < .05$.

emerged for the P3b (main effect, $p = .257$, $\eta^2 = .02$; interaction effects, all $ps > .05$, $\eta^2 < .01$). However, for the extreme group variable a significant main effect of NFC was observed, $F(1, 36) = 4.98$, $p = .032$, $\eta^2 = .12$. As depicted in Figure 3B (right panel), high-NFC participants showed larger P3b factor scores ($M = 0.22$, $SEM = 0.03$) and thus larger P3b amplitudes than did low-NFC participants ($M = 0.12$, $SEM = 0.03$). This effect remained unaffected if the influence of both the early P3a and late P3a component were statistically controlled, $F(1, 34) = 4.69$, $p = .037$, $\eta^2 = .12$. Furthermore, none of the other personality dimensions significantly affected the P3b component (all $ps > .05$).

Behavioral data. The mean target count of the sample was close to 100% of the correct target number

($M = 98.2\%$, $SD = 3.23$). No effect of NFC on target count was observed ($p > .05$).

Discussion

The findings of Study 2 replicate and extend those of Study 1. Again, for the extreme group variable, an NFC main effect was found for the late P3a component, which also occurred for the continuous variable. This confirms the result of Study 1 and points out that NFC is linked to bottom-up-driven processes of attentional resource allocation in response to contextually novel events. Contrary to Study 1, high-NFC individuals also showed marginally higher scores in the early P3a component. However, this effect disappeared after statistically controlling for the variance shared with the adjacent components even after PCA-decorrelation. Thus, regarding the P3a subcomponents, the results indicate that NFC-specific influences mainly occur on temporarily later levels of involuntary or automatic information processing as reflected by the late P3a component.

Given that high-NFC individuals prefer effortful cognitive activities and typically tend to invest a higher degree of cognitive resources in task-related activities than do low-NFC individuals (e.g., Cacioppo & Petty, 1982; Day et al., 2007; Dornic et al., 1991), we particularly hypothesized NFC-specific differences in the P3b amplitude, which reflects a voluntary, top-down-driven attentional resource allocation. Due to the lack of NFC-specific influence on the P3b component in Study 1, we increased the cognitive demand by doubling the total number of stimuli in Study 2 so that attention to the presented stimuli had to be maintained throughout a prolonged period of time. This modification was made because NFC-specific differences in voluntary attentional resource allocation should particularly occur in cognitive-demanding situations. Indeed, significant differences between high- and low-NFC participants were observed for the P3b component in Study 2, with larger amplitudes for high-NFC individuals. It has to be noted that this effect was not due to less noise in the ERP because of a higher number of trials available for averaging and, hence, a more precise estimate of P3 component factor scores because the SNR of Study 2 did not differ from that of Study 1. Moreover, the behavioral paradigm conducted prior to the novelty oddball task (see Procedure) also may have increased the overall demand and may have further fostered NFC-specific differences. In addition, the sample of Study 2 was more homogeneous because it exclusively included undergraduate psychology students. It may be speculated that a higher homogeneity might hold constant factors that also influence the ERPs. Nevertheless, the differences in design and sample are rather subtle. Furthermore, the

effect was only observed within the extreme group design, which may overestimate effect size (McClelland & Judd, 1993). Therefore, although being consistent with NFC theory and behavioral evidence, the finding of greater voluntary attentional resource allocation by high-NFC individuals as indicated by target-related P3b amplitude requires further investigation.

To ensure that the observed effects are specific to NFC, we also investigated the influence of the personality factors Novelty Seeking and Sensation Seeking as well as Extraversion, Openness to Experience, and Conscientiousness that are either assumed to affect the P3 components or show association with NFC. Only Conscientiousness was significantly related to the late P3a, which at first sight provides convergent validity because NFC and Conscientiousness are behaviorally linked. However, the influence of NFC on involuntary attention allocation indicated by the late P3a amplitude appears to be substantially greater because the observed effect of Conscientiousness disappeared after entering NFC into the same ANOVA model, whereas the influence of NFC remained significant. This, in turn, provides evidence of the discriminant validity of NFC-specific resource allocation.

GENERAL DISCUSSION

The present research examined cognitive mechanisms underlying NFC-specific variation in information processing using a neurocognitive paradigm that provides an objective online measure to investigate effects of cognitive effort or effortful processing concerning NFC. We found first evidence for event-related brain potentials that relate NFC to voluntary but also to automatic or involuntary attention allocation.

In both Studies 1 and 2, high-NFC participants showed substantially larger late P3a amplitudes in response to rare, contextually novel stimuli than did their low-NFC counterparts. It has been suggested that the P3a mirrors an evaluative, precategorical aspect of orienting leading to processes that solidify the formation of a new, modality-specific stimulus representation (Cycowicz & Friedman, 1997). This allows individuals to quickly interpret events of potential behavioral or biological significance and to rapidly prepare an accurate response (Daffner et al., 2003; Friedman et al., 2001). This also may result in a more effective stimulus encoding with positive effects on later recall because the magnitude of the P3a response is positively associated with the degree to which the information is subsequently remembered (Fabiani & Donchin, 1995; Ranganath & Rainer, 2003). Given that NFC is positively correlated with both recall and recognition (e.g., Kardash & Noel, 2000; Peltier & Schibrowsky,

1994), we speculate that a more efficient attentional orienting or monitoring system as indicated by P3a amplitude at least partially contributes to the elaborated processing style and the better recall commonly reported for high-NFC individuals (e.g., Cacioppo et al., 1996). In addition, because individuals high in NFC are described as active exploring agents in the constructing of meaning who enjoy dealing with ambiguity, it is reasonable to assume that their thinking also includes monitoring and regulating their mentation in a metacognitive sense (Perkins, Tishman, Ritchhart, Donis, & Andrade, 2000).

In Study 2, but not in Study 1, we found significantly larger P3b (or target P3) amplitudes for high- relative to low-NFC individuals, indicating greater attentional resource allocation with regard to voluntary, top-down-driven target processing. In general, the P3b is assumed to reflect a neuronal correlate of stimulus evaluation and categorization and covaries with the amount of attention paid to the stimulus (e.g., Kok, 2001; Polich, 2003). This can be interpreted as reflecting the degree of cognitive involvement (Debener et al., 2002) and the intensity of processing (Kok, 2001), respectively. Attentional engagement also is believed to be associated with the updating of the actual stimulus representation held in working memory that provides a link between attention and memory reflected by the P3b (Donchin & Coles, 1988; Polich, 2007). Besides the P3a association, the results of Study 2 also provide evidence of NFC-specific differences in voluntary, top-down-driven processes of resource allocation and point to the usefulness of the P3b as an objective marker of cognitive effort or “cognitive efficiency” (Polich, 2004).

Limitations and Future Directions

Our results, particularly those of Study 2, indicate that individuals high in NFC allocate a greater amount of attention to both contextually novel as well as target relevant information. This suggests a general tendency of a deeper elaboration and consolidation of information because both the P3a and P3b amplitude have been linked to effective encoding and better memory performance (Fabiani & Donchin, 1995; Polich, 2004, 2007). However, silently counting the target tones as was the behavioral task in the current design was not appropriate to meaningfully examine such performance differences between NFC groups. Therefore, future research should include additional performance measures (e.g., recall or RT-measures) that can be linked to NFC-specific variations in the P3a and P3b amplitudes.

A related potential limitation of the present work refers to the task difficulty of the paradigm because the target stimuli were comparatively easy to discriminate from the standard tones and the novel sounds. Consequently,

NFC-specific differences in target processing as indicated by P3b variation may have been limited to Study 2, where cognitive demand was enhanced by doubling the number of presented stimuli. Therefore, future research also should enhance the cognitive requirements of the paradigm, for example, by reducing perceptual distinctiveness between standard and target tones. Moreover, increasing task demand also might contribute to clarify whether NFC-specific differences in bottom-up-driven processing of novel stimuli remain unaffected if participants are highly engaged in target-specific processing or whether they solely occurred because high-NFC individuals were searching for a cognitive challenge provided by the novel events in the otherwise fatiguing setting. Because of the interplay between attention and memory highlighted in the P3b, future research also should control for (working) memory processes that might help to further specify the relation between NFC and the P3b.

Concluding Remarks

As proposed by NFC theory, differential cognitive resource allocation (or cognitive effort) is assumed to be the core aspect of interindividual differences in cognitive motivation and a major determinant of NFC-specific performance differences in a variety of tasks. Using a neurobehavioral paradigm and analyzing event-related potentials, the present research provides evidence for neurophysiological correlates of attentional resource allocation relating to NFC. Our results suggest that NFC-specific processing is not limited to voluntary or executive aspects of resource allocation but also relies on automatic, bottom-up-driven attention allocation. We argue that these involuntary processes may facilitate conscious interpretation, elaboration, and memory consolidation of information and in conjunction with voluntary processes contribute to cognitive efficiency. We hope that this specification of the conceptualization of NFC will provide a better understanding of individual differences in cognitive motivation and cognitive effort as the key mediator of the elaboration likelihood of information.

NOTE

1. In the current context, principal components analysis (PCA) was preferred over independent component analysis (ICA) for signal decomposition and decorrelation. Although as compared to orthogonality of the PCA-derived components, ICA results in independency of the decomposed signals of electroencephalogram (EEG, i.e., also takes into account higher-order correlations), ICA of EEG recordings from a restricted number of channels, such as 32 in the present study, often results in insufficient unmixing of underlying independent signals and/or in several independent components for event-related brain potentials (ERPs) such as the P3a and the P3b, with the latter issue requiring component clustering, which is prone to subjective decisions. PCA, on the other hand, results in unambiguous and singular

parameters for a given ERP for every participant, and using varimax rotation, at least orthogonality is warranted, which is especially important when examining differential effects of a variable on spatiotemporally partly overlapping ERP components such as the early and late P3a and the P3b.

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